Lumen: An Open, Differentiable, GPU-Parallel Environment for Wall-Safe Endovascular AI

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Abstract

Endovascular autonomy needs simulation environments that expose the signals an agent must optimize: branch selection, target reach, wall interaction, image observations, and reproducible safety outcomes. Lumen is an Apache-2.0 endovascular AI environment built around procedural vascular cases, tube-intrinsic contact, GPU-parallel Newton/Warp execution, Gymnasium-compatible navigation tasks, synthetic fluoroscopy, luminal image generation, and replayable computer-vision datasets. This preprint describes the public release and reports a launch benchmark on lumen-bench/2, where a deterministic forward baseline achieved 100% safe success across tube, stenotic, and branch-navigation tasks while retaining explicit wall-penetration accounting. Relative to CathSim and SOFA/BeamAdapter-derived navigation stacks, Lumen emphasizes benchmarked wall safety, multi-modal labels, and reproducible dataset artifacts from the same simulator state.

1 Motivation

Autonomous catheter and guidewire navigation is now an active research direction for robotic endovascular intervention, with reinforcement learning, imitation learning, inverse reinforcement learning, and expert-trajectory learning explored in simulated and physical settings [4, 5, 1, 2, 3, 6]. CathSim demonstrated the value of an open endovascular simulator for autonomous catheterization research by pairing high-fidelity robot/device simulation with reinforcement-learning tasks [1, 2]. SOFA and BeamAdapter provide a separate lineage: finite-element flexible-device modeling for interactive interventional-radiology simulation and mechanical-thrombectomy navigation studies [7, 8, 5].

Lumen is aimed at the benchmark layer between these lines of work. It exposes procedural vascular cases, wall-state semantics, image observations, label generation, and safety-scored leaderboard outputs through a single public stack. The core design decision is that target reach and wall-safe target reach are separate outcomes. A policy that reaches a branch while breaching the wall is not treated as equivalent to a policy that reaches the same target without unsafe interaction.

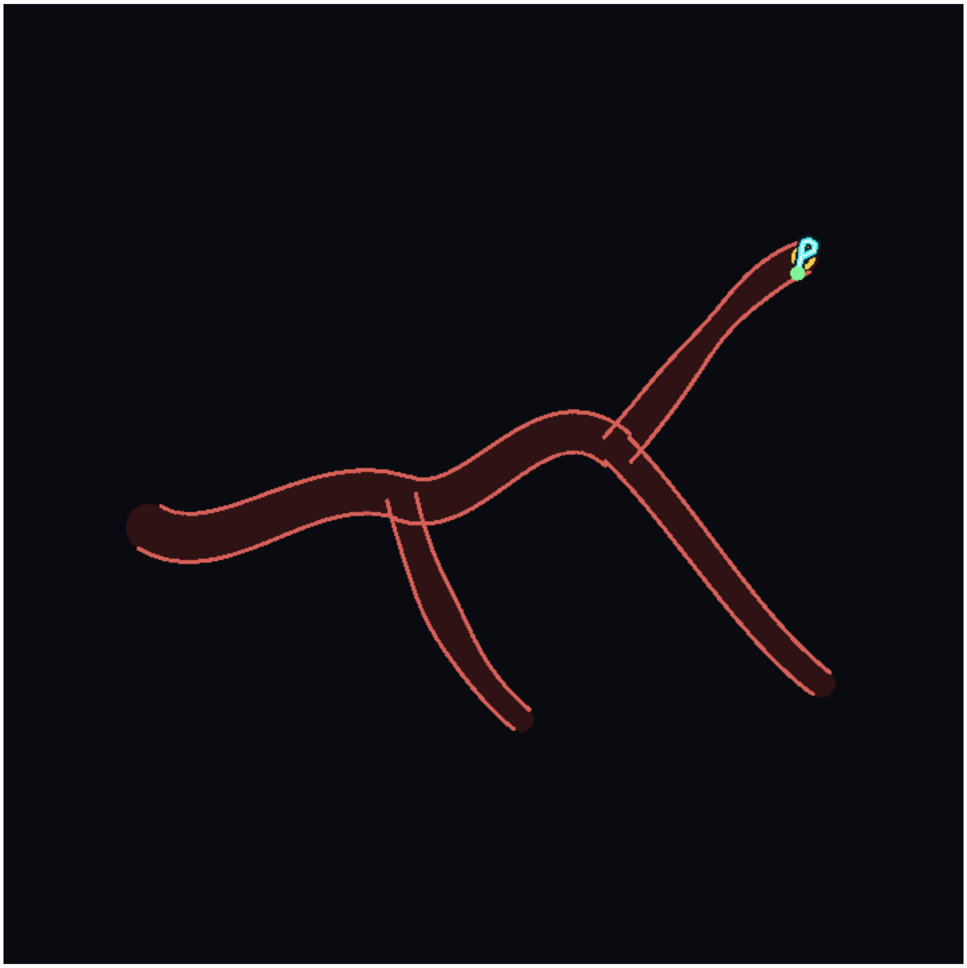


Figure 1: Simulator-native Lumen navigation capture from the current implementation. The same procedural vascular tree drives policy state, wall-safety accounting, rendered observations, and replay metadata.

2 System Overview

Lumen is organized around five connected surfaces:

1. **Procedural vascular cases.** Tube, stenotic, tortuous, and branching vascular cases are generated as replayable scenes, enabling deterministic reruns and benchmark submissions.

2. **Tube-intrinsic contact and wall state.** Device progress is represented along the vessel manifold, with route progress, contact, penetration, torsion, and friction surfaced as environment state.

3. **GPU-parallel execution.** Newton/Warp-style kernels support high-throughput differentiable simulation paths for calibration and batched environment work [13].

4. **Observation and dataset generation.** Fluoroscopy, luminal RGB, segmentation masks, keypoints, device masks, and detector-noise variants are generated from the same scene state.

5. **Gymnasium-compatible RL tasks.** Navigation tasks are registered through the Gymnasium interface, allowing standard reinforcement-learning tooling to treat Lumen as a conventional environment [14, 15].

This arrangement follows the successful pattern of general RL benchmarks [14, 15], physics-backed control suites [22, 23], and medical-simulation frameworks [7, 9, 10], while specializing state, observation, and scoring surfaces for endovascular device navigation.

3 Observation Layer

Lumen emits state observations and image observations from the same case. A single scene can produce biplanar digitally reconstructed radiographs, masks for vessel and device geometry, labeled keypoints, noisy fluoroscopy, and a luminal RGB camera view. These outputs support reinforcement learning, computer-vision pretraining, perception debugging, and dataset construction.

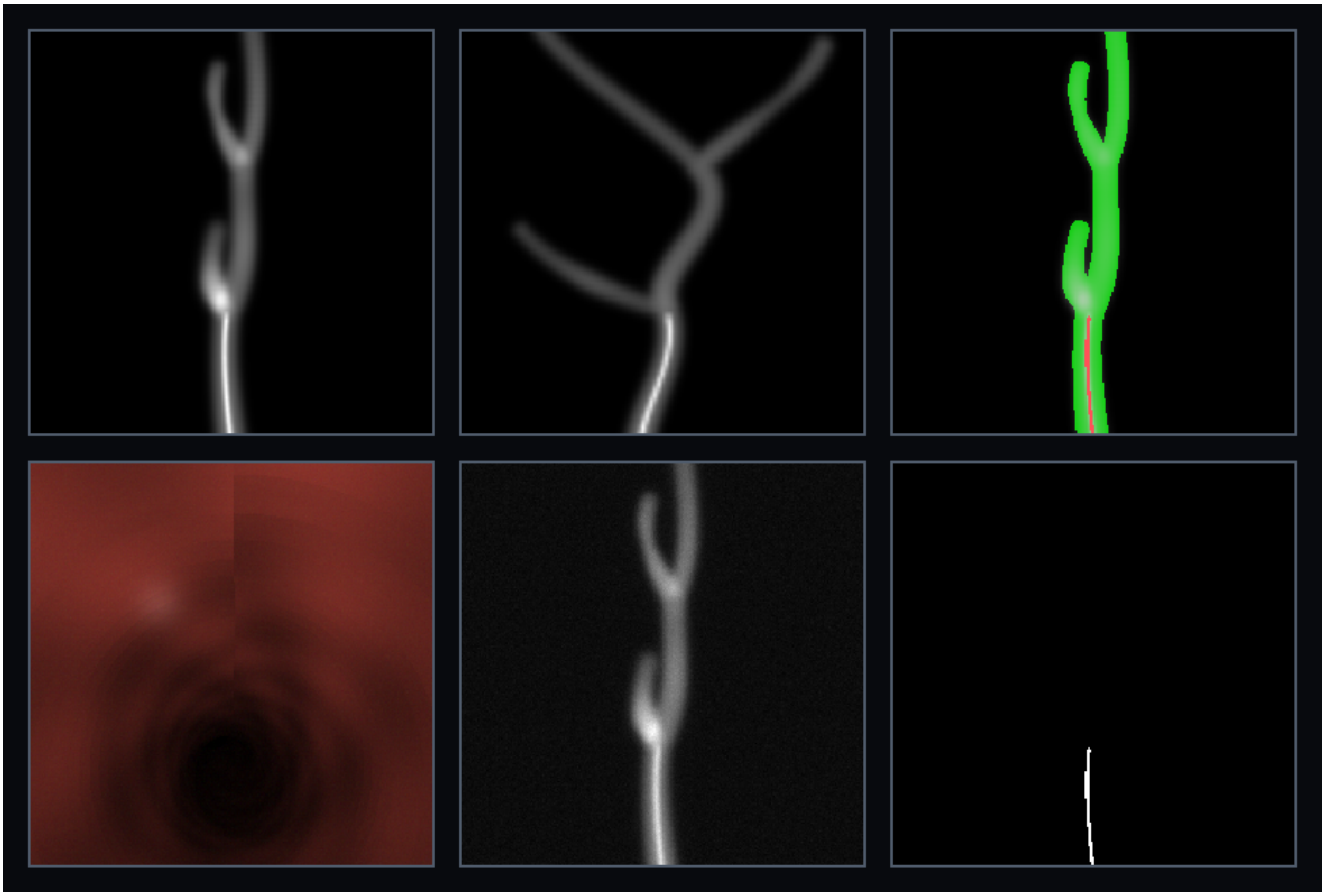


Figure 2: Multi-modal scene outputs from one Lumen case: biplanar fluoroscopy, vessel/device labels, luminal RGB, detector noise, and masks. The same geometry drives control state and image data.

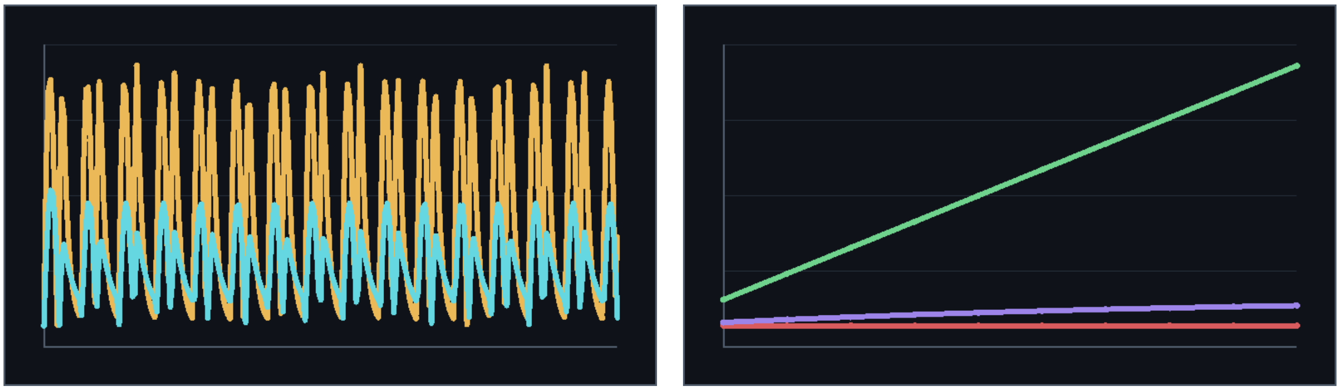
The dataset workflow is exposed as command-line tooling for capture, validation, indexing, and train/validation/test splitting. Episode sidecars make label, mask, and replay metadata auditable before the dataset is consumed by a model.

4 Physics and Device Modules

Lumen includes reduced-order modules for phenomena that should influence policy behavior:

- deformable-wall semantics inspired by arterial-wall mechanics and anisotropic constitutive modeling [11, 12];

- contact and penetration metrics reported in environment state;
- tortuosity, route progress, and branch-target scoring;
- flow-diverter effects and aneurysm inflow traces;
- clot-field interaction, stentriever retrieval, fragmentation, and damage bookkeeping;
- synthetic fluoroscopy and luminal rendering for image-observation agents.



(a) Aneurysm inflow before and after a flow-diverter state change.

(b) Stentriever retrieval, captured clot, and damage traces.

Figure 3: Simulator-native advanced module outputs from the current implementation. These are raw Lumen captures, not website screenshots or generated promotional imagery.

The simulator is not only a renderer around a centerline. It is a benchmark substrate where contact, flow, clot/device state, image formation, and replay metadata can be captured together and reproduced from case metadata.

5 Benchmark Protocol and Results

The launch benchmark, `lumen-bench/2`, evaluates three fixed procedural tasks: `nav_tube`, `nav_stenotic`, and `nav_tree_branch`. Each task is run for five seeded episodes. The scorecard ranks safe target reach before raw target reach, then wall safety, then return. The safety threshold for the launch scorecard is a maximum wall penetration of 0.3 simulator units.

Figure 4: Navigation rollouts in branching and tortuous vessels. Lumen reports route progress, target reach, wall penetration, and safety status so benchmark results can distinguish safe target reach from unsafe reach.

Table 1: Launch benchmark result for the deterministic `forward-baseline` policy on `lumen-bench/2`. The scorecard is included with the release artifacts.

Task	Tier	Episodes	Safe success	Unsafe success	Mean steps	Max pen.
`nav_tube`	Easy	5	1.00	0.00	18.6	0.000
`nav_stenotic`	Medium	5	1.00	0.00	19.8	0.000
`nav_tree_branch`	Hard	5	1.00	0.00	54.0	0.178
Overall	–	15	1.00	0.00	–	0.178

The forward baseline is intentionally simple: it advances the proximal end at full rate. It is useful as a reproducibility check and a floor for trained policies, not as a ceiling. Standard reinforcement-learning algorithms such as PPO and SAC can be applied through the Gymnasium interface [16, 17, 18], while leaderboard submissions preserve per-task scorecards for comparison.

6 Comparison to CathSim and SOFA/BeamAdapter

CathSim, SOFA/BeamAdapter, and Lumen occupy related but distinct positions in the open endovascular-simulation landscape. CathSim is the closest open autonomous-catheterization benchmark reference, with

MuJoCo-based robot/device simulation, force sensing, and reinforcement-learning tasks [1, 2]. SOFA/BeamAdapter is a mature flexible-instrument simulation surface for catheters, guidewires, and coils, and has been used as the basis for SOFA mechanical-thrombectomy navigation studies [7, 8, 5]. Lumen is designed around a compact public benchmark where wall safety, image labels, and replayable datasets are first-class outputs.

Table 2: Capability positioning of Lumen against commonly cited open endovascular-simulation references.

Surface	Lumen	CathSim	SOFA/BeamAdapter
Primary target	Wall-safe AI benchmark and data generation	Autonomous catheterization simulator and RL tasks	Flexible catheter/guidewire mechanics
RL interface	Gymnasium tasks plus portable scorecards	PPO/SAC-style RL evaluation reported	Used by navigation studies; no common leaderboard
Safety metric	Safe success separated from unsafe reach	Contact/force simulation surface	SOFA-scene contact mechanics
Imaging and labels	Fluoro, luminal RGB, masks, keypoints, noisy detector output	Visual simulation and segmentation surfaces	Scene-dependent rendering and mechanics
Geometry	Replayable tube, stenotic, tortuous, and branch cases	Aorta/robot simulation assets	User-authored SOFA scenes and vascular models
Release emphasis	Reproducible benchmark artifacts, CV-ready captures, Apache-2.0	Open autonomous catheterization simulator	Open mechanics framework and plugin ecosystem

The practical contribution is that Lumen packages wall-safety scoring, procedural task reproducibility, multi-modal labels, and replay artifacts into one public benchmark release that can be inspected outside the RL loop.

7 Availability

The public repository is github.com/SeldingerMed/seldinger-lumen. The release includes the benchmark scorecard, simulator-native figures, launch video, source LaTeX, and generated PDF.

8 Conclusion

Lumen provides an open endovascular AI environment where vessel navigation, wall safety, synthetic imaging, labels, and replayable benchmark artifacts are emitted from the same stack. The launch benchmark establishes a reproducible baseline for safe target reach; the comparison with CathSim and SOFA/BeamAdapter clarifies the role of Lumen as a safety-scored, dataset-ready benchmark rather than another isolated viewer or mechanics demo.

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